

MDS211 Nervous System — SA1 Mock Exam

Lectures 1–16 · 77 Questions · Single Best Answer

This mirrors the REAL SA1 exam structure from your official schedule: 77 questions distributed across Lectures 1–16 in proportion to actual lecture hours (15 hours total), matching the same weighting your real Summative Exam 1 uses. Every question is pulled from my verified, fact-checked question bank (the same one behind “The Professor’s Gauntlet”), not the original error-prone past-paper archive — so the questions AND the answer key here are both accurate. Time yourself: the real SA1 is 2.5 hours for 77 questions across Lec 1–16 plus other content, so budget roughly 2 minutes per question here.

Lecture 1 — Introduction & Organization of the CNS (5 questions)

1. Concerning the parts of the Central Nervous System, which of the following items is NOT included?
 - A. Pineal gland
 - B. Basal ganglia
 - C. Fourth ventricle
 - D. Trigeminal ganglion
 - E. Posterior pituitary gland
2. When you wear a ring for a long time, you stop noticing it. Which receptor’s rapid adaptation is responsible?
 - A. Meissner’s corpuscle
 - B. Pacinian corpuscle
 - C. Free nerve ending
 - D. Merkel disc
 - E. Ruffini ending
3. Which of the following is NOT a division of the peripheral nervous system?
 - A. Somatic nervous system
 - B. Autonomic nervous system
 - C. Enteric nervous system
 - D. Central nervous system
 - E. Cranial and spinal nerves

† original question, written from lecture content
4. A layer-by-layer dissection proceeds from the skull inward. Put the meningeal layers in the correct order.
 - A. Dura → Pia → Arachnoid
 - B. Arachnoid → Dura → Pia
 - C. Dura → Arachnoid → Pia
 - D. Pia → Arachnoid → Dura
 - E. Pia → Dura → Arachnoid

† original question, written from lecture content
5. CSF produced in the lateral ventricles must pass through the foramen of Monro to reach the:
 - A. Fourth ventricle
 - B. Third ventricle
 - C. Cerebral aqueduct
 - D. Subarachnoid space directly
 - E. Central canal

† original question, written from lecture content

Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

6. Which of the following compounds is transferred from astrocyte to neuron and used for ATP production during starvation?
- A. Fumarate
 - B. Citrate
 - C. Pyruvate
 - D. Succinyl-CoA
 - E. Lactate
7. Myelin sheath wrapping nerve axons within the basal ganglia (a CNS structure) is produced by which cell?
- A. Astrocyte
 - B. Schwann cell
 - C. Satellite cell
 - D. Oligodendrocyte
 - E. Microglial cell
8. Which of the following is the glutamate transporter responsible for uptake into astrocytes?
- A. MCT
 - B. DCT
 - C. IAATS
 - D. TCT
 - E. EAATs

Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

9. A 40-year-old male needs an urgent hand suture and is given lidocaine as local anesthesia. What is its mechanism of action?
- A. Activation of ligand-gated potassium channels
 - B. Inhibition of nicotinic acetylcholine receptors
 - C. Inhibition of voltage-gated sodium channels
 - D. Inhibition of acetylcholine release
 - E. Activation of muscarinic acetylcholine receptors
10. Hyperpolarization of the postsynaptic membrane in response to glycine or GABA-A receptor activation is produced by the opening of which channel?
- A. Na⁺ channels
 - B. K⁺ channels only
 - C. Cl⁻ channels
 - D. Ca²⁺ channels
 - E. Nonspecific cation channels

- 11.** A 35-year-old male receives a lidocaine injection before incision and drainage of a buttock abscess. Which best describes lidocaine's mechanism?
- A. Blocks voltage-gated K⁺ channels
 - B. Blocks ligand-gated Ca²⁺ channels
 - C. Blocks voltage-gated Na⁺ channels
 - D. Stimulates ligand-gated Na⁺ channels
 - E. Blocks Mg²⁺ channels
- 12.** A patient develops muscle weakness that worsens through the afternoon and evening but is nearly absent in the morning. Which pathology is most consistent?
- A. Guillain-Barré syndrome
 - B. Myasthenia gravis
 - C. Multiple sclerosis
 - D. Duchenne muscular dystrophy
 - E. Amyotrophic lateral sclerosis
- 13.** What is the major consumer of energy (ATP) in the resting brain?
- A. DNA replication in neurons
 - B. Restoring ionic gradients via the Na⁺/K⁺-ATPase
 - C. Myelin synthesis in oligodendrocytes
 - D. Muscle contraction at the periphery
 - E. Protein degradation in the soma

Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

- 14.** Which of the following is INCORRECT regarding neuronal metabolism?
- A. Neurons preferentially utilize aerobic oxidative phosphorylation for ATP production
 - B. Astrocytes preferentially perform anaerobic glycolysis, producing lactate
 - C. Neurons store large amounts of glycogen for backup fuel
 - D. Neurons contain LDH-1, which favors pyruvate formation for entry into the TCA cycle
 - E. Astrocytes contain LDH-5, which favors lactate formation from pyruvate
- 15.** What enzyme hydrolyzes glutamine to glutamate inside the presynaptic neuron, regenerating the neurotransmitter pool?
- A. Glutamine synthetase
 - B. Glutaminase
 - C. Glutamate decarboxylase
 - D. GABA transaminase
 - E. Monoamine oxidase
- 16.** Excess extracellular glutamate causes excitotoxic neuronal injury primarily through excessive intracellular accumulation of which ion?
- A. Na⁺
 - B. K⁺
 - C. Ca²⁺
 - D. Cl⁻
 - E. Mg²⁺

17. A 58-year-old man with cirrhosis develops confusion and drowsiness with markedly elevated serum ammonia (hepatic encephalopathy). Which cell is primarily responsible for detoxifying the ammonia in the brain, and how?

- A. Neurons, via glutaminase enzyme activity
- B. Astrocytes, via glutamine synthetase
- C. Microglia, via ammonia phagocytosis
- D. Oligodendrocytes, via myelin synthesis
- E. Ependymal cells, via CSF filtration

† original question, written from lecture content

18. In the same hepatic encephalopathy scenario, why does excessive glutamine accumulation in astrocytes become harmful rather than simply protective?

- A. Glutamine is directly neurotoxic to channels
- B. Glutamine acts as an osmolyte, drawing water into cells
- C. Glutamine inhibits the Na⁺/K⁺-ATPase directly
- D. Glutamine converts back to ammonia spontaneously
- E. Glutamine has no downstream metabolic effect

† original question, written from lecture content

Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)

19. Which embryonic structure gives rise to the substantia nigra?

- A. Metencephalon
- B. Myelencephalon
- C. Telencephalon
- D. Diencephalon
- E. Mesencephalon

20. The optic cup is an evagination (outgrowth) of which structure?

- A. Surface ectoderm of the face
- B. Neuroectoderm of the diencephalon
- C. Mesoderm of the head region
- D. Neural crest cell migration
- E. Endoderm of the pharynx

21. A meningocele is characterized by protrusion of which structure(s) through a vertebral/cranial defect?

- A. Meninges and CSF only, no neural tissue
- B. Neural tissue only, with no meninges
- C. Bone of the vertebral arch only
- D. Meninges plus spinal cord tissue
- E. Skin and subcutaneous tissue only

22. Sustained high intracranial pressure classically produces which triad of findings (Cushing's triad)?

- A. Hypertension, bradycardia, irregular breathing
- B. Hypotension, tachycardia, rapid breathing
- C. Fever, skin rash, light sensitivity
- D. Hypertension, tachycardia, hyperventilation
- E. Hypotension, bradycardia, apnea episodes

23. What does the ventral (inferior) part of the optic vesicle, at the point where it connects to the forebrain, give rise to?
- A. The developing lens placode
 - B. The retinal (choroid) fissure and hyaloid vessels
 - C. The extraocular muscle primordia region
 - D. The developing eyelid fold tissue
 - E. The lacrimal gland anlage tissue

Lecture 7 — Spinal Cord & Blood Supply (5 questions)

24. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which finding is expected on the SAME (right) side at and below L2?
- A. Loss of pain and temperature sensation
 - B. Loss of proprioception, vibration, and motor power
 - C. Complete anesthesia of the limb
 - D. Isolated loss of pain sensation only
 - E. No deficit on the affected side
25. A 65-year-old man has lost vibration, position sense, and joint discrimination in the legs, with preserved pain and temperature sensation. Which tract/pathway is most likely damaged?
- A. Lateral spinothalamic tract damage
 - B. Dorsal column-medial lemniscus pathway damage
 - C. Anterior spinothalamic tract damage
 - D. Corticospinal tract damage only
 - E. Spinocerebellar tract damage
26. Nuchal rigidity (stiff neck) in a patient is a compensatory sign classically seen with which condition?
- A. Simple tension-type headache pattern
 - B. Meningeal irritation from bleeding or infection
 - C. Peripheral nerve neuropathy pattern
 - D. Migraine without aura pattern
 - E. Isolated Bell's palsy pattern
27. An injury damages the LATERAL HORN of the spinal cord gray matter (present at T1-L2/L3). Which function is most directly lost?
- A. Somatic motor function to skeletal muscle
 - B. Preganglionic sympathetic output
 - C. Dorsal column sensation
 - D. Spinothalamic sensation
 - E. Cerebellar coordination
28. All of the following statements concerning the anterior spinal artery are correct EXCEPT:
- A. Supplies the anterior two-thirds of the cord
 - B. Arises from paired vertebral artery branches
 - C. Territory includes corticospinal and spinothalamic tracts
 - D. Occlusion spares the dorsal columns
 - E. Reinforced only at cervical level, nowhere else

Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

29. Which functional component is found in the hypoglossal nucleus?
- A. GSA
 - B. SVA
 - C. GVA
 - D. SVE
 - E. GSE
30. According to the organization of the PNS, which statement below is INCORRECT?
- A. Spinal nerves are formed by the union of anterior and posterior nerve roots
 - B. The trigeminal ganglion contains pseudounipolar sensory neurons for face/oral cavity sensation
 - C. There is only one functional component in a peripheral nerve
 - D. Secretomotor functions from the brainstem are transmitted via CN III, VII, IX, X
 - E. There are 12 pairs of cranial nerves, and only CN III-XII emerge from the brainstem
31. Collections of nerve cell bodies located inside the peripheral nervous system are called:
- A. Tracts
 - B. Nerve fibers
 - C. Ganglia
 - D. Nerves
 - E. Nuclei

Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

32. An MRI shows a tumor compressing the right tuberculum cinereum at the closed (caudal) medulla. Which nerve's nucleus is most directly compressed?
- A. Trigeminal nerve nucleus/tract
 - B. Facial nerve nucleus
 - C. Glossopharyngeal nerve nucleus
 - D. Vagus nerve nucleus
 - E. Accessory nerve nucleus
33. Which reflex test best represents intact function of the medulla oblongata?
- A. Gag reflex
 - B. Knee-jerk reflex
 - C. Babinski reflex
 - D. Pupillary light reflex
 - E. Corneal reflex
34. A 20-year-old man has loss of muscle power on the LEFT side of his body plus strabismus and ptosis of the RIGHT eye. Where is the lesion?
- A. Left cerebral cortex lesion pattern
 - B. Right pons lesion pattern
 - C. Rostral midbrain, right side (Weber)
 - D. Left cerebellum lesion pattern
 - E. Right medulla lesion pattern

- 35.** Which neuronal pathway performs voluntary control of body and limb movements?
- A. Reticulospinal tract
 - B. Corticopontine tract
 - C. Posterior column-medial lemniscus pathway
 - D. Hypothalamospinal tract
 - E. Lateral corticospinal (pyramidal) tract
- 36.** Which CNS structure lies at approximately the same horizontal (axial) level as the tentorial notch/incisura, useful for localizing herniation syndromes?
- A. Medulla oblongata
 - B. Midbrain
 - C. Pons
 - D. Cervical spinal cord
 - E. Cerebellar tonsils
- 37.** A 32-year-old man has loss of muscle power and a positive Babinski sign on the LEFT side, with right-sided tongue deviation on protrusion. Where is the lesion?
- A. Left cerebral cortex lesion pattern
 - B. Right medial medulla syndrome
 - C. Left cerebellum lesion pattern
 - D. Right pons lesion pattern
 - E. Left midbrain lesion pattern
- 38.** A patient has a left pupil larger than the right, plus ptosis and an eye deviated 'down and out.' Which structure is most likely compressed?
- A. Left optic nerve lesion
 - B. Left oculomotor nerve (CN III) lesion
 - C. Right abducens nerve lesion
 - D. Left trochlear nerve lesion only
 - E. Left trigeminal nerve lesion
- 39.** A patient has loss of pain and temperature sensation on the CONTRALATERAL body plus an IPSILATERAL facial pain/temperature deficit, plus ipsilateral ataxia, Horner's syndrome, and hoarseness. This constellation is:
- A. Weber syndrome pattern
 - B. Lateral medullary (Wallenberg) syndrome
 - C. Medial medullary syndrome pattern
 - D. Locked-in syndrome pattern
 - E. Benedikt syndrome pattern
- 40.** Which of the following correctly pairs a reticular formation column with its function?
- A. Median column — arousal and attention
 - B. Lateral column — controls eye movements only
 - C. Medial column — has no known function
 - D. All three columns are functionally identical
 - E. Median column — descending reflexes only

41. Fibers from which reticular formation structure synapse with the motor nuclei of CN V and CN VII to help mediate brainstem reflexes like the jaw jerk and blink?
- A. Median column (raphe nuclei) directly
 - B. Medial column, via short interneuronal links
 - C. Lateral column exclusively
 - D. Locus coeruleus acting alone
 - E. Substantia nigra acting alone

Lecture 11 — Special Sense Organs & Its Development (8 questions)

42. The inner ear (membranous labyrinth) develops embryologically from which structure?
- A. First pharyngeal cleft
 - B. Otic placode
 - C. First pharyngeal pouch
 - D. Neural crest cells
 - E. Surface ectoderm overlying the eye

† original question, written from lecture content

43. The external auditory meatus develops from which embryonic structure?
- A. First pharyngeal pouch
 - B. First pharyngeal cleft
 - C. Otic vesicle
 - D. Second pharyngeal arch
 - E. First pharyngeal arch mesenchyme

† original question, written from lecture content

44. The malleus and incus (middle ear ossicles) are derived embryologically from which cartilage?
- A. Reichert's cartilage (2nd arch)
 - B. Meckel's cartilage (1st arch)
 - C. Third arch cartilage
 - D. Otic capsule cartilage
 - E. Laryngeal cartilage

† original question, written from lecture content

45. Which cell type in the olfactory epithelium serves as the stem cell population for continuous regeneration of olfactory receptor neurons?
- A. Sustentacular cells
 - B. Bowman's gland cells
 - C. Basal cells
 - D. Olfactory receptor neurons themselves
 - E. Schwann cells

† original question, written from lecture content

46. The nasal placodes are induced by signals from which adjacent embryonic structure?
- A. The developing forebrain tissue
 - B. The developing heart tube region
 - C. The nearby otic vesicle
 - D. The pharyngeal pouches nearby
 - E. The notochord below it

† original question, written from lecture content

47. Which taste bud cell type is directly responsible for chemoreceptive taste transduction?

- A. Basal (Type IV) supporting cells
- B. Dark cells of the taste bud
- C. Gustatory taste receptor cells
- D. Supporting cells only, no transduction
- E. Merkel cells of the epithelium

† original question, written from lecture content

48. Which special sense epithelium is uniquely capable of continuous neurogenesis (producing new sensory NEURONS, not just supporting cells) throughout adult life?

- A. Cochlea (organ of Corti)
- B. Retina
- C. Olfactory epithelium
- D. Vestibular maculae
- E. Taste buds

† original question, written from lecture content

49. The cochlear duct (scala media) contains which fluid, and how does its ionic composition differ from the adjacent scalae?

- A. Perilymph, similar to CSF
- B. Endolymph, unusually high in K⁺
- C. Blood plasma directly bathing hair cells
- D. Pure water with no dissolved ions
- E. Same fluid as scala vestibuli/tympani

† original question, written from lecture content

Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

50. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which neuronal tract/fasciculus passes through the anterior white commissure at the level of decussation?

- A. Corticospinal tract crossing point
- B. Dorsal column, uncrossed pathway
- C. Lateral spinothalamic tract, crosses early
- D. Spinocerebellar tract, mostly uncrossed
- E. Rubrospinal tract crossing point

51. A patient has loss of proprioception on the LEFT side and loss of pain/temperature on the RIGHT side of the body below a spinal lesion. This pattern is most consistent with:

- A. Complete cord transection pattern
- B. Left Brown-Séquard hemisection pattern
- C. Anterior spinal artery syndrome pattern
- D. Posterior spinal artery syndrome pattern
- E. Peripheral neuropathy pattern

52. Which tract lesion classically gives a positive Romberg test (patient sways/falls with eyes closed but is stable with eyes open)?
- A. Corticospinal tract
 - B. Dorsal column-medial lemniscus pathway (proprioception loss compensated visually)
 - C. Lateral spinothalamic tract
 - D. Cerebellar vermis (this instead causes a Romberg-NEGATIVE, i.e., always-unsteady, truncal ataxia)
 - E. Optic nerve
53. Which of the following conditions primarily affects the PNS rather than the CNS?
- A. Multiple sclerosis
 - B. Guillain-Barré syndrome
 - C. Stroke (ischemic)
 - D. Parkinson's disease
 - E. Amyotrophic lateral sclerosis (upper motor neuron component)
54. A patient has right-sided pain and temperature loss below T12, and left-sided pain and temperature loss below T10 (staggered levels). What does this pattern suggest about the lesion?
- A. A single precise transverse hemisection
 - B. An expanding syrinx hitting crossing fibers unevenly
 - C. A pure peripheral nerve lesion
 - D. Normal variant anatomy, no pathology
 - E. A pure dorsal column lesion
55. The ganglia housing first-order taste neurons are located in which structures?
- A. Geniculate, petrosal, nodose (CN VII, IX, X)
 - B. Superior, ciliary, and otic ganglia
 - C. Stellate, geniculate, and nodose ganglia
 - D. Trigeminal, geniculate, nodose ganglia
 - E. Geniculate, spiral, pterygopalatine ganglia
56. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?
- A. Satellite cells
 - B. Schwann cells
 - C. Basal cells
 - D. Stem keratinocytes
 - E. Merkel cells
57. The gustatory (taste) pathway includes all of the following EXCEPT:
- A. Solitary nucleus (rostral/gustatory part)
 - B. VPM nucleus of the thalamus
 - C. Geniculate/petrosal/nodose ganglia
 - D. Insula and frontal operculum (primary gustatory cortex)
 - E. Trapezoid body

Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

58. A woman presents with depressive symptoms and suicidal thoughts. Which neurotransmitter is classically implicated as abnormal in this case?

- A. Glutamate
- B. Acetylcholine
- C. Serotonin
- D. GABA
- E. Dopamine

59. What is the mechanism of action of botulinum toxin?

- A. Increased Na⁺ channel activity presynaptically
- B. Blocks ACh receptor postsynaptically
- C. Cleaves SNARE proteins, blocking vesicle fusion
- D. Blocks Ca²⁺ channels presynaptically
- E. Increased reuptake of acetylcholine

60. An 80-year-old man has progressive short-term memory loss and word-finding difficulty, diagnosed with Alzheimer disease. Which statement about its pathogenesis is INCORRECT?

- A. Atrophy of gyri in the cerebral cortex
- B. Formation of amyloid plaques
- C. Formation of neurofibrillary tangles
- D. Increased activity of beta-secretase enzyme
- E. Blocking of dopamine reuptake transporters

61. A patient arrives with continuous generalized tonic-clonic seizures for 15 minutes (status epilepticus). What is the most appropriate FIRST-line treatment?

- A. Diazepam, a benzodiazepine
- B. Valproate, an anticonvulsant
- C. Phenytoin, an anticonvulsant
- D. Fluoxetine, an antidepressant
- E. Memantine, an NMDA antagonist

62. The summation of EPSPs arriving from numerous different presynaptic fibers converging onto the same postsynaptic neuron at roughly the same moment is called:

- A. Spatial summation
- B. Temporal summation
- C. Presynaptic inhibition
- D. Long-term potentiation
- E. Presynaptic facilitation

63. A 20-year-old college student has depressive symptoms including sleep disturbance and anorexia for 1 month, with no prior mood disorder history. What is the mechanism of the drug most appropriate for this patient (an SSRI)?

- A. Stimulation of GABA receptors directly
- B. Inhibition of dopamine receptors
- C. Inhibition of the serotonin transporter (SERT)
- D. Inhibition of acetylcholinesterase enzyme
- E. Stimulation of norepinephrine release

- 64.** A woman presents with progressive muscle weakness, dysphagia, and ptosis that worsens with repeated blinking. Which finding is most likely present?
- A. Ryanodine receptor autoantibody
 - B. Anti-presynaptic Ca^{2+} channel antibody
 - C. Anti-nicotinic ACh receptor antibody
 - D. Dopamine receptor inhibition
 - E. GABA receptor stimulation directly
- 65.** A 45-year-old man has a deep gardening wound, then develops jaw stiffness (lockjaw) and muscle spasms. Which mechanism best explains the responsible toxin's action?
- A. Inhibits ACh release at the NMJ directly
 - B. Autoantibody against nicotinic ACh receptors
 - C. Inhibits Ca^{2+} channels postsynaptically
 - D. Blocks inhibitory interneurons in the cord
- 66.** A chemotherapy patient has nausea and vomiting. Which antiemetic receptor target is most appropriate?
- A. 5-HT₃ receptor, e.g. ondansetron
 - B. Dopamine receptor blockade
 - C. Muscarinic receptor blockade
 - D. Nicotinic receptor blockade
 - E. Adrenergic receptor blockade
- 67.** Which of the following agents does NOT stimulate nociceptors (i.e., is not itself an inflammatory/pain-sensitizing mediator)?
- A. Prostaglandin
 - B. K^{+}
 - C. Substance P
 - D. Histamine
 - E. Vibration

Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

- 68.** A patient exhibits a near-vision reflex defect. Which of the following is NOT a normal characteristic of the near vision reflex?
- A. Medial deviation (convergence) of both eyes
 - B. Levator palpebrae contraction, lid elevation
 - C. Smaller pupil diameter, constriction
 - D. Increased lens convexity, accommodation
- 69.** A patient has a tumor compressing the optic chiasm. Which visual field defect results?
- A. Left homonymous hemianopia pattern
 - B. Right homonymous hemianopia pattern
 - C. Bitemporal heteronymous hemianopia
 - D. Binasal heteronymous hemianopia

- 70.** Which part of the retina has NO photoreceptor cells at all?
- A. Macula lutea region
 - B. Fovea centralis region
 - C. Retina near the ora serrata
 - D. Optic disc, the physiologic blind spot
 - E. Paramacular retinal field
- 71.** What structure in the eye is primarily responsible for absorbing stray light, reducing internal reflection/glare?
- A. The pigmented choroid layer
 - B. The neural retina layer
 - C. The corneal epithelium
 - D. The outer scleral coat
- 72.** Which condition is indicated by an intraocular pressure (IOP) greater than 25 mmHg?
- A. Cataract
 - B. Retinal detachment
 - C. Macular degeneration
 - D. Glaucoma
- 73.** Which structure is found at the level of the rostral pons?
- A. Cerebral aqueduct region
 - B. Abducens nucleus specifically
 - C. Area postrema specifically
 - D. Motor and principal nuclei of CN V
 - E. Inferior vestibular nucleus
- 74.** A patient has facial nerve damage resulting in hyperacusis (abnormally loud perception of sound). Which structure's function is most likely interrupted?
- A. Trapezoid body damage
 - B. Superior olivary complex damage
 - C. Tensor tympani reflex loss
 - D. Stapedius reflex loss, CN VII lesion
 - E. Lateral lemniscus damage
- 75.** Which vestibular nucleus sends axons directly to the oculomotor nucleus (contributing to the vestibulo-ocular reflex)?
- A. Medial vestibular nucleus
 - B. Lateral vestibular nucleus
 - C. Superior vestibular nucleus
 - D. Inferior vestibular nucleus
- 76.** Which structure releases (dissipates) the pressure wave of sound energy out of the inner ear, allowing continued fluid movement within the cochlea?
- A. Eardrum (tympanic membrane)
 - B. Oval window
 - C. Round window
 - D. Helicotrema
 - E. Scala vascularis (not a real structure)

77. Which statement regarding sound stimulation and cochlear transduction is TRUE?

- A. Endolymph vibration through Reissner's membrane
- B. Outer hair cells generate the main afferent signal
- C. Inner hair cells signal via spiral ganglion fibers
- D. High frequency stimulates the cochlear apex

Answer Key & Explanations

Lectures 1–16 · 77 Questions

Lecture 1 — Introduction & Organization of the CNS (5 questions)

Q1. Concerning the parts of the Central Nervous System, which of the following items is NOT included?

✓ **Correct: D. Trigeminal ganglion**

The trigeminal ganglion sits outside the brain proper (in Meckel's cave) and houses first-order sensory neuron cell bodies — that makes it a PNS structure, not CNS.

✗ **A is wrong:** The pineal gland is neuroectoderm-derived and physically part of the diencephalon roof — it counts as CNS even though it's an endocrine gland.

✗ **E is wrong:** The posterior pituitary (neurohypophysis) is a direct downgrowth of the hypothalamus/diencephalon, so it's CNS tissue despite sitting outside the cranial vault proper.

TIP: Classic trick: examiners mix in endocrine-sounding structures (pineal, posterior pituitary) that are secretly CNS because of their embryonic origin from the neural tube.

Q2. When you wear a ring for a long time, you stop noticing it. Which receptor's rapid adaptation is responsible?

✓ **Correct: B. Pacinian corpuscle**

Pacinian corpuscles are the fastest-adapting (phasic) mechanoreceptors, firing only at the onset/offset of a stimulus — perfect for 'you stop feeling constant pressure.'

✗ **A is wrong:** Meissner's corpuscles are also rapidly adapting, but they encode light touch/flutter at the fingertips, not sustained deep pressure like a ring.

✗ **D is wrong:** Merkel discs are slowly adapting (tonic) — they'd keep firing the whole time you wore the ring, which is the opposite of what the question describes.

TIP: Sort receptors into two adaptation buckets first: SLOW (Merkel, Ruffini) = keep firing, sense duration/pressure; FAST (Meissner, Pacinian) = fire once, sense change/vibration.

Q3. Which of the following is NOT a division of the peripheral nervous system?

✓ **Correct: D. Central nervous system**

By definition the CNS (brain + spinal cord) is the other half of the nervous system, not a subdivision of the PNS.

✗ **C is wrong:** The enteric nervous system is often forgotten, but it's a genuine PNS division — it can function semi-autonomously even when vagal/sympathetic input is cut.

Q4. A layer-by-layer dissection proceeds from the skull inward. Put the meningeal layers in the correct order.

✓ **Correct: C. Dura → Arachnoid → Pia**

From outside in: Dura (tough, periosteal+meningeal layers) → Arachnoid (web-like, subarachnoid space beneath it holds CSF) → Pia (hugs the brain surface, follows every gyrus/sulcus).

✗ **D is wrong:** This reverses the order completely — pia is the innermost layer, directly adherent to neural tissue, not the outermost.

TIP: Mnemonic: 'Dr. AP' — Dura, Arachnoid, Pia, outside to in. The subdural space is a potential space (between dura & arachnoid); the subarachnoid space is real and full of CSF.

Q5. CSF produced in the lateral ventricles must pass through the foramen of Monro to reach the:

✓ **Correct: B. Third ventricle**

The interventricular foramen (of Monro) connects each lateral ventricle to the third ventricle — the next stop on the CSF circuit.

✗ **A is wrong:** The fourth ventricle is two stops away — CSF must still cross the cerebral aqueduct (of Sylvius) from the third ventricle before reaching the fourth.

TIP: Know the full CSF circuit cold: Lateral ventricles → (foramen of Monro) → 3rd ventricle → (cerebral aqueduct) → 4th ventricle → (foramina of Luschka/Magendie) → subarachnoid space → arachnoid granulations → venous sinuses.

Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

Q6. Which of the following compounds is transferred from astrocyte to neuron and used for ATP production during starvation?

✓ **Correct: E. Lactate**

The astrocyte-neuron lactate shuttle exports lactate (via MCT) from glycolytic astrocytes to fuel neuronal oxidative metabolism, especially when demand is high or glucose delivery is limited.

✗ **C is wrong:** *Pyruvate is the end-product of glycolysis inside a cell, but it isn't the molecule classically shuttled between astrocyte and neuron — lactate (which astrocytes preferentially generate via LDH5) is.*

TIP: 'Astrocyte-neuron LACTATE shuttle' — if you remember only one word from this topic, remember 'lactate,' not 'pyruvate' or 'glucose.'

Q7. Myelin sheath wrapping nerve axons within the basal ganglia (a CNS structure) is produced by which cell?

✓ **Correct: D. Oligodendrocyte**

Oligodendrocytes myelinate CNS axons, and each oligodendrocyte can myelinate multiple axon segments simultaneously.

✗ **B is wrong:** *Schwann cells myelinate PNS axons only (one Schwann cell = one internode on one axon) — a very common mix-up since both cells 'do the same job' in different compartments.*

TIP: CNS myelin = Oligodendrocyte (multiple axons at once). PNS myelin = Schwann cell (one axon, one cell). If the question mentions basal ganglia, cortex, spinal cord tracts = CNS = oligodendrocyte.

Q8. Which of the following is the glutamate transporter responsible for uptake into astrocytes?

✓ **Correct: E. EAATs**

Excitatory Amino-Acid Transporters (EAATs, especially EAAT-2/GLT-1) on astrocytic membranes clear synaptic glutamate to prevent excitotoxicity.

✗ **A is wrong:** *MCT (monocarboxylate transporter) moves lactate/pyruvate/ketone bodies, not glutamate — easy to confuse since both transporters live on astrocytes and both are exam favorites.*

Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

Q9. A 40-year-old male needs an urgent hand suture and is given lidocaine as local anesthesia. What is its mechanism of action?

✓ **Correct: C. Inhibition of voltage-gated sodium channels**

Lidocaine (an amide local anesthetic) blocks voltage-gated Na⁺ channels from the intracellular side, preventing the rising phase of the action potential and thus blocking nociceptive transmission.

✗ **D is wrong:** *That describes botulinum toxin's mechanism (blocking ACh vesicle fusion), not a local anesthetic — a classic mechanism mix-up between two totally different drug classes.*

*TIP: Amide local anesthetics all end in 'i-caine' with two i's (lido**ca**lne, bupivac**al**ne) vs ester types with one i (procaine, tetracaine) — useful for allergy history questions too, since esters cross-react via PABA metabolites and amides generally don't.*

Q10. Hyperpolarization of the postsynaptic membrane in response to glycine or GABA-A receptor activation is produced by the opening of which...

✓ **Correct: C. Cl⁻ channels**

GABA-A and glycine receptors are both ionotropic ligand-gated Cl⁻ channels; Cl⁻ influx (down its electrochemical gradient in most mature neurons) hyperpolarizes/stabilizes the membrane — classic fast inhibition.

✗ *B is wrong: K⁺ channels do contribute to some inhibitory postsynaptic potentials (e.g., via GABA-B, a metabotropic receptor), but the FAST, ionotropic GABA-A/glycine response specifically is a Cl⁻ channel, not K⁺.*

Q11. A 35-year-old male receives a lidocaine injection before incision and drainage of a buttock abscess. Which best describes lidocaine's...

✓ **Correct: C. Blocks voltage-gated Na⁺ channels**

Same core fact as any lidocaine question: it's a voltage-gated Na⁺ channel blocker, preventing depolarization and action potential propagation in nociceptive fibers.

TIP: If you see 'lidocaine,' 'bupivacaine,' or 'procaine' anywhere in a stem, your first instinct should always be 'voltage-gated Na⁺ channel blockade' unless the question is specifically asking about something else (e.g., allergy class).

Q12. A patient develops muscle weakness that worsens through the afternoon and evening but is nearly absent in the morning. Which pathology...

✓ **Correct: B. Myasthenia gravis**

Myasthenia gravis produces fatigable weakness that accumulates over the course of the day/with use as postsynaptic ACh receptors become progressively less available, then partially recovers with rest overnight.

Q13. What is the major consumer of energy (ATP) in the resting brain?

✓ **Correct: B. Restoring ionic gradients via the Na⁺/K⁺-ATPase**

The largest share of brain ATP consumption goes to running the Na⁺/K⁺-ATPase, restoring the Na⁺/K⁺ gradients dissipated by action potentials and synaptic currents — the brain is metabolically expensive largely because of this pump.

Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

Q14. Which of the following is INCORRECT regarding neuronal metabolism?

✓ **Correct: C. Neurons store large amounts of glycogen for backup fuel**

Neurons store essentially NO glycogen — astrocytes are the brain's glycogen reservoir, not neurons; this is a key reason neurons depend on a steady glucose/lactate supply from surrounding glia.

TIP: Glycogen storage in the CNS = astrocyte job, not neuron. This single fact underlies why the astrocyte-neuron lactate shuttle exists in the first place — neurons need a buffer they don't keep themselves.

Q15. What enzyme hydrolyzes glutamine to glutamate inside the presynaptic neuron, regenerating the neurotransmitter pool?

✓ **Correct: B. Glutaminase**

Glutaminase (phosphate-activated glutaminase, PAG) converts glutamine back into glutamate inside neurons, completing the glutamate-glutamine cycle after astrocytes have converted the neuron's released glutamate into (non-excitotoxic) glutamine.

✗ *A is wrong: Glutamine synthetase runs the OPPOSITE reaction (glutamate → glutamine) and is located in ASTROCYTES, not neurons — a very commonly reversed pair.*

✗ *C is wrong: Glutamate decarboxylase converts glutamate into GABA (in GABAergic neurons), a completely different fate for glutamate, not glutamine regeneration.*

TIP: Glutamine SYNTHETASE (astrocyte, makes glutamine) vs GlutaminASE (neuron, breaks down glutamine back to glutamate). One letter of difference in the name, opposite direction, opposite cell — the exam's favorite pair to swap.

Q16. Excess extracellular glutamate causes excitotoxic neuronal injury primarily through excessive intracellular accumulation of which ion?

✓ **Correct: C. Ca^{2+}**

Overactivation of NMDA (and AMPA) receptors lets excessive Ca^{2+} flood into the postsynaptic neuron, triggering destructive enzyme cascades (proteases, lipases, free radical formation) — the hallmark of excitotoxicity seen in stroke/ischemia and hyperammonemia.

Q17. A 58-year-old man with cirrhosis develops confusion and drowsiness with markedly elevated serum ammonia (hepatic encephalopathy). Which...

✓ **Correct: B. Astrocytes, via glutamine synthetase**

Astrocytes are the brain's main ammonia detoxifiers: glutamine synthetase combines ammonia with glutamate to form glutamine, trapping the nitrogen in a non-toxic form — but this consumes large amounts of glutamate and ATP, and excess glutamine itself is what drives astrocyte swelling in hepatic encephalopathy.

✗ *A is wrong: Neurons run glutaminase, the opposite reaction (breaking glutamine down) — they don't have significant ammonia-detoxifying capacity; that job belongs to astrocytic glutamine synthetase.*

TIP: This is a real exam-style integrative case: liver failure → ↑ammonia → astrocytes overwork glutamine synthetase → glutamine accumulates intracellularly → osmotic astrocyte swelling → cerebral edema/confusion. Multi-step 'why' chains like this are exactly what a good prof will test.

Q18. In the same hepatic encephalopathy scenario, why does excessive glutamine accumulation in astrocytes become harmful rather than simply...

✓ **Correct: B. Glutamine acts as an osmolyte, drawing water into cells**

Glutamine is osmotically active; when astrocytes are forced to make large amounts of it to buffer ammonia, intracellular osmolarity rises and water follows, causing astrocyte (and eventually brain) swelling — a major contributor to the cerebral edema seen in acute hepatic encephalopathy.

Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)

Q19. Which embryonic structure gives rise to the substantia nigra?

✓ **Correct: E. Mesencephalon**

The substantia nigra is a midbrain (mesencephalon) structure — it develops in place as part of the mesencephalic vesicle, not by migrating from another region.

✗ *D is wrong: The diencephalon gives rise to the thalamus, hypothalamus, and related structures — it's immediately rostral to the mesencephalon, an easy neighboring-vesicle mix-up.*

TIP: Memorize the 5 secondary vesicles in order, rostral to caudal: Telencephalon, Diencephalon, Mesencephalon, Metencephalon, Myelencephalon — 'Tell Danny My Mom's Mad.' Substantia nigra = mesencephalon, no exceptions.

Q20. The optic cup is an evagination (outgrowth) of which structure?

✓ **Correct: B. Neuroectoderm of the diencephalon**

The optic vesicle (itself an outpouching of the diencephalic neuroectoderm) invaginates to form the double-walled optic cup — the retina's neuroectodermal origin.

✗ *A is wrong: Surface ectoderm forms the LENS placode/vesicle (and later the corneal epithelium) — a separate embryonic contribution that fuses with, but is distinct from, the optic cup.*

TIP: Two different germ-layer contributions build the eye: Neuroectoderm (diencephalon) → optic vesicle → optic cup → retina/optic nerve. Surface ectoderm → lens placode → lens, and → corneal epithelium.

Q21. A meningocele is characterized by protrusion of which structure(s) through a vertebral/cranial defect?

✓ **Correct: A. Meninges and CSF only, no neural tissue**

A meningocele is a herniation of meninges (and the CSF they contain) through a bony defect, WITHOUT neural tissue — generally a milder lesion than a myelomeningocele.

✗ *D is wrong: That combination (meninges + spinal cord) describes a myelomeningocele, the more severe form — a very frequently tested distinction.*

Q22. Sustained high intracranial pressure classically produces which triad of findings (Cushing's triad)?

✓ **Correct: A. Hypertension, bradycardia, irregular breathing**

Cushing's triad (rising ICP) = hypertension (widened pulse pressure), reflex bradycardia (baroreceptor response), and irregular respirations — a late, ominous sign of impending herniation.

TIP: Cushing's triad direction is easy to get backwards under exam pressure: BP goes UP, HR goes DOWN. Remember it's the body trying to maintain cerebral perfusion pressure against the rising ICP, which paradoxically triggers a baroreceptor-mediated bradycardia.

Q23. What does the ventral (inferior) part of the optic vesicle, at the point where it connects to the forebrain, give rise to?

✓ **Correct: B. The retinal (choroid) fissure and hyaloid vessels**

The ventral surface of the optic vesicle/stalk/cup develops a groove — the retinal (choroid) fissure — allowing the hyaloid artery/vein to enter and supply the developing inner eye structures before it normally closes.

Lecture 7 — Spinal Cord & Blood Supply (5 questions)

Q24. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which finding is expected on the SAME (right)...

✓ **Correct: B. Loss of proprioception, vibration, and motor power**

Hemisection knocks out the IPSILATERAL dorsal columns (proprioception/vibration) and the ipsilateral lateral corticospinal tract (spastic weakness) below the lesion, because these tracts haven't crossed yet at that spinal level; pain/temperature loss is CONTRALATERAL because the spinothalamic tract has already crossed.

✗ *A is wrong: Pain and temperature loss is CONTRALATERAL (opposite side), not ipsilateral — the spinothalamic tract crosses near its level of entry, so damage is felt on the opposite side of the body below the lesion.*

TIP: Brown-Séquard's core logic: dorsal columns + corticospinal tract = ipsilateral loss (they cross in the MEDULLA, high up, so below a cord lesion they're still uncrossed on that side). Spinothalamic tract = contralateral loss (it crosses within 1-2 SEGMENTS of entering the cord, so it's already crossed by the time it reaches the lesion level).

Q25. A 65-year-old man has lost vibration, position sense, and joint discrimination in the legs, with preserved pain and temperature...

✓ **Correct: B. Dorsal column-medial lemniscus pathway damage**

Vibration, proprioception, and fine touch/discrimination are dorsal column-medial lemniscus functions; sparing of pain/temperature (spinothalamic) points to a selective dorsal column lesion — classically seen in conditions like tabes dorsalis (tertiary syphilis) or vitamin B12 deficiency (subacute combined degeneration).

TIP: This 'dissociated sensory loss' pattern (one modality out, another spared) is a classic clinical reasoning trick — always map which specific tract carries which specific modality, because these questions test whether you can localize a lesion from the symptom pattern alone.

Q26. Nuchal rigidity (stiff neck) in a patient is a compensatory sign classically seen with which condition?

✓ **Correct: B. Meningeal irritation from bleeding or infection**

Neck stiffness limiting flexion is a protective reflex against stretching inflamed/irritated meninges, classically seen in meningitis or subarachnoid hemorrhage — tested clinically with Kernig's and Brudzinski's signs.

Q27. An injury damages the LATERAL HORN of the spinal cord gray matter (present at T1-L2/L3). Which function is most directly lost?

✓ **Correct: B. Preganglionic sympathetic output**

The lateral horn (intermediolateral cell column), present only at thoracolumbar levels, contains preganglionic sympathetic neuron cell bodies — its destruction disrupts sympathetic outflow from that segment.

✗ *A is wrong: Somatic motor neurons are in the VENTRAL (anterior) horn, a completely different gray matter region present at ALL spinal levels, not just T1-L2/L3.*

TIP: Lateral horn = ONLY present T1-L2/L3 (sympathetic preganglionic neurons) plus S2-S4 has an analogous parasympathetic cell column. If a question mentions 'lateral horn,' it's testing autonomic (usually sympathetic), not somatic motor, function.

Q28. All of the following statements concerning the anterior spinal artery are correct EXCEPT:

✓ **Correct: E. Reinforced only at cervical level, nowhere else**

This is FALSE (hence the correct EXCEPT answer): the anterior spinal artery is reinforced along its length by multiple segmental radicular arteries, most famously the large artery of Adamkiewicz in the lower thoracic/upper lumbar region — reinforcement is not limited to the cervical level.

TIP: Artery of Adamkiewicz (major radicular artery, usually T9-T12) is a classic exam name-drop — its damage (e.g., during aortic surgery) causes anterior spinal artery syndrome affecting the lower cord.

Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

Q29. Which functional component is found in the hypoglossal nucleus?

✓ **Correct: E. GSE**

Repeated core fact: hypoglossal nerve (CN XII) = pure motor to tongue muscles = GSE.

Q30. According to the organization of the PNS, which statement below is INCORRECT?

✓ **Correct: C. There is only one functional component in a peripheral nerve**

This is the FALSE statement (hence correct as the 'INCORRECT' answer): peripheral nerves are typically MIXED, containing multiple functional components simultaneously (e.g., GSA + GVE + SVE in a single cranial nerve like the facial nerve).

Q31. Collections of nerve cell bodies located inside the peripheral nervous system are called:

✓ **Correct: C. Ganglia**

Ganglia = PNS cell body clusters. The CNS equivalent term is 'nucleus'; a bundle of axons is a 'tract' in the CNS or a 'nerve' in the PNS.

✗ *E is wrong: 'Nuclei' is the correct term, but specifically for CNS cell body collections — a PNS collection is called a ganglion, not a nucleus.*

TIP: Cell bodies: CNS = nucleus, PNS = ganglion. Axon bundles: CNS = tract, PNS = nerve. Four terms, two locations, memorize the grid.

Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

Q32. An MRI shows a tumor compressing the right tuberculum cinereum at the closed (caudal) medulla. Which nerve's nucleus is most directly...

✓ **Correct: A. Trigeminal nerve nucleus/tract**

The tuberculum cinereum overlies the spinal trigeminal nucleus and tract in the closed medulla — a classic surface landmark for underlying trigeminal sensory structures.

TIP: 'Tuberculum cinereum' = trigeminal spinal nucleus landmark. If you see this phrase anywhere in a stem, your brain should immediately go to CN V (trigeminal), specifically its spinal sensory nucleus for pain/temperature of the face.

Q33. Which reflex test best represents intact function of the medulla oblongata?

✓ Correct: A. Gag reflex

The gag reflex depends on CN IX (afferent) and CN X (efferent), both with nuclei in the medulla — a classic 'brainstem is intact' bedside test, especially relevant in assessing brain death/coma.

✗ E is wrong: The corneal reflex tests CN V (afferent) and CN VII (efferent), both PONTINE nuclei — one level rostral to the medulla, a common mix-up with the gag reflex.

TIP: Brainstem reflex ladder (rostral to caudal): Pupillary light reflex (midbrain, CN II/III) → Corneal reflex (pons, CN V/VII) → Gag reflex (medulla, CN IX/X). Testing this ladder in order is exactly how brain death exams localize the level of brainstem dysfunction.

Q34. A 20-year-old man has loss of muscle power on the LEFT side of his body plus strabismus and ptosis of the RIGHT eye. Where is the lesion?

✓ Correct: C. Rostral midbrain, right side (Weber)

This is a classic Weber syndrome (medial midbrain syndrome): a right-sided lesion damages the exiting right CN III fascicles (ipsilateral oculomotor palsy — ptosis, strabismus) AND the right cerebral peduncle/corticospinal fibers before they've crossed (causing LEFT, contralateral, hemiplegia).

TIP: 'Crossed' brainstem syndromes (cranial nerve deficit on ONE side, body weakness on the OTHER) are the single highest-yield brainstem localization pattern — the cranial nerve tells you WHICH side and WHICH level; the body deficit is always contralateral because corticospinal fibers haven't crossed yet at that point.

Q35. Which neuronal pathway performs voluntary control of body and limb movements?

✓ Correct: E. Lateral corticospinal (pyramidal) tract

The lateral corticospinal tract is THE primary pathway for voluntary, skilled limb movement, descending from motor cortex, crossing at the pyramidal decussation, and synapsing on lower motor neurons.

✗ A is wrong: The reticulospinal tract is involved in more automatic postural/gross movement control (an extrapyramidal pathway) — it is NOT the pathway for fine voluntary control, a common trap distractor.

Q36. Which CNS structure lies at approximately the same horizontal (axial) level as the tentorial notch/incisura, useful for localizing...

✓ Correct: B. Midbrain

The midbrain passes through the tentorial incisura (the opening in the tentorium cerebelli) — this is exactly why UNCAL (transtentorial) herniation compresses the midbrain first, producing the classic ipsilateral CN III palsy (blown pupil) of a rapidly expanding supratentorial mass.

Q37. A 32-year-old man has loss of muscle power and a positive Babinski sign on the LEFT side, with right-sided tongue deviation on...

✓ Correct: B. Right medial medulla syndrome

Medial medullary syndrome (occlusion of the anterior spinal artery/vertebral artery branches) damages the ipsilateral (right) hypoglossal nerve fascicles — causing tongue deviation TOWARD the lesion (right) on protrusion — plus the pyramid (contralateral, left, corticospinal weakness/Babinski) and medial lemniscus (contralateral dorsal column loss).

TIP: Tongue deviates TOWARD the side of a CN XII (hypoglossal) lesion because the genioglossus muscle on the weak side can't push the tongue away — combine that with contralateral body weakness and you've localized a crossed medial medullary syndrome instantly.

Q38. A patient has a left pupil larger than the right, plus ptosis and an eye deviated 'down and out.' Which structure is most likely compressed?

✓ **Correct: B. Left oculomotor nerve (CN III) lesion**

A dilated ('blown'), unreactive pupil plus ptosis plus a down-and-out eye is the classic complete CN III palsy triad — the parasympathetic fibers controlling pupil constriction run on the OUTSIDE of the nerve, making them especially vulnerable to compression (as opposed to ischemic CN III palsies, which tend to spare the pupil).

TIP: 'Surgical' (compressive) CN III palsy = pupil involved (blown, dilated) because the parasympathetic fibers sit peripherally and get compressed first. 'Medical' (ischemic, e.g., diabetic) CN III palsy = pupil SPARED, because the vasa nervorum ischemia hits the central fibers first, sparing the peripheral parasympathetic fibers. This pupil-sparing distinction is one of the most classic tested contrasts in all of neurology.

Q39. A patient has loss of pain and temperature sensation on the CONTRALATERAL body plus an IPSILATERAL facial pain/temperature deficit, plus...

✓ **Correct: B. Lateral medullary (Wallenberg) syndrome**

Wallenberg syndrome (lateral medullary syndrome, classically from PICA or vertebral artery occlusion) hits the spinal trigeminal nucleus/tract (ipsilateral facial pain/temp loss), the lateral spinothalamic tract (contralateral body pain/temp loss), the inferior cerebellar peduncle (ipsilateral ataxia), descending sympathetics (ipsilateral Horner's), and nucleus ambiguus (hoarseness/dysphagia) — a huge, high-yield symptom cluster.

TIP: Wallenberg = 'everything BUT motor' (the pyramids/corticospinal tract are anteromedial, spared in this LATERAL medullary infarct) — that absence of weakness is itself a diagnostic clue, since most other brainstem strokes DO cause some limb weakness.

Q40. Which of the following correctly pairs a reticular formation column with its function?

✓ **Correct: A. Median column — arousal and attention**

The median column (raphe nuclei) sends ascending serotonergic projections toward the cortex/thalamus, important for arousal, attention, and mood regulation.

TIP: Three reticular formation columns, three jobs: Median (raphe) = ascending, arousal/attention/mood. Medial (central) = intrinsic brainstem reflexes (eye movement, swallowing). Lateral = descending, muscle tone/respiration/blood pressure control.

Q41. Fibers from which reticular formation structure synapse with the motor nuclei of CN V and CN VII to help mediate brainstem reflexes like...

✓ **Correct: B. Medial column, via short interneuronal links**

The medial (central) reticular column provides the local, intrinsic interneuronal connections linking sensory input to motor cranial nerve nuclei for brainstem reflexes (eye movements, swallowing, and other coordinated reflex arcs).

Lecture 11 — Special Sense Organs & Its Development (8 questions)

Q42. The inner ear (membranous labyrinth) develops embryologically from which structure?

✓ **Correct: B. Otic placode**

The otic placode invaginates to form the otic vesicle (otocyst), which gives rise to the entire membranous labyrinth.

TIP: Ear develops from THREE different embryonic pouches/clefts/placodes: external ear (1st cleft), middle ear (1st pouch), inner ear (otic placode, neuroectoderm) — three separate structures fusing into one organ, a favorite integrative exam theme.

Q43. The external auditory meatus develops from which embryonic structure?

✓ **Correct: B. First pharyngeal cleft**

The first pharyngeal cleft forms the external auditory meatus, while the corresponding first pharyngeal pouch forms the tympanic cavity and auditory tube.

✗ A is wrong: Pouches (endoderm-lined) form the MIDDLE ear cavity; clefts (ectoderm-lined) form the EXTERNAL ear canal — an easy pouch/cleft swap.

Q44. The malleus and incus (middle ear ossicles) are derived embryologically from which cartilage?

✓ **Correct: B. Meckel's cartilage (1st arch)**

Meckel's cartilage (1st pharyngeal arch) gives rise to the malleus and incus; Reichert's cartilage (2nd arch) gives rise to the stapes.

TIP: 1st arch (Meckel's) → malleus + incus. 2nd arch (Reichert's) → stapes. This mismatch (2 ossicles from arch 1, only 1 from arch 2) is a favorite 'which arch gives which ossicle' trick.

Q45. Which cell type in the olfactory epithelium serves as the stem cell population for continuous regeneration of olfactory receptor neurons?

✓ **Correct: C. Basal cells**

Basal cells are the progenitor/stem cell population of the olfactory epithelium, continuously regenerating olfactory receptor neurons throughout life.

TIP: Basal cells are the universal stem cell answer for BOTH olfactory epithelium AND taste buds — an easy fact to double up on two different exam questions.

Q46. The nasal placodes are induced by signals from which adjacent embryonic structure?

✓ **Correct: A. The developing forebrain tissue**

Nasal placodes form on the frontonasal prominence under inductive signals from the underlying forebrain — another example (alongside the eye) of the CNS directly inducing an adjacent special sense organ.

Q47. Which taste bud cell type is directly responsible for chemoreceptive taste transduction?

✓ **Correct: C. Gustatory taste receptor cells**

Taste receptor cells (Type II/III) within the taste bud are the chemosensory cells that transduce taste stimuli via microvilli projecting into the taste pore.

Q48. Which special sense epithelium is uniquely capable of continuous neurogenesis (producing new sensory NEURONS, not just supporting cells)...

✓ **Correct: C. Olfactory epithelium**

Olfactory receptor neurons are continuously regenerated from basal stem cells — a distinctive feature not shared by cochlear hair cells or retinal photoreceptors, which do not regenerate in adult humans.

X E is wrong: Taste bud cells DO turn over regularly, but the taste receptor cells are specialized epithelial cells, not true neurons — the olfactory epithelium is unique in regenerating actual bipolar sensory NEURONS with axons that must find their way back into the CNS (olfactory bulb).

Q49. The cochlear duct (scala media) contains which fluid, and how does its ionic composition differ from the adjacent scalae?

✓ **Correct: B. Endolymph, unusually high in K⁺**

The cochlear duct (scala media) is filled with endolymph, notable for its high K⁺ / low Na⁺ composition (unlike typical extracellular fluid) — this unusual ionic gradient, maintained by the stria vascularis, is what actually drives the receptor (transduction) current when hair cell channels open.

TIP: Endolymph (scala media) = high K⁺, resembles intracellular fluid. Perilymph (scala vestibuli & scala tympani) = low K⁺/high Na⁺, resembles typical extracellular fluid/CSF. This K⁺-driven transduction current (not Na⁺, unusually) is a favorite physiology trick question.

Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

Q50. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which neuronal tract/fasciculus passes through...

✓ **Correct: C. Lateral spinothalamic tract, crosses early**

Spinothalamic second-order neurons cross the midline via the anterior (ventral) white commissure, just anterior to the central canal, within 1-2 segments of entering the cord — this is the anatomical crossing point tested repeatedly in Brown-Séquard-type questions.

Q51. A patient has loss of proprioception on the LEFT side and loss of pain/temperature on the RIGHT side of the body below a spinal lesion....

✓ **Correct: B. Left Brown-Séquard hemisection pattern**

This exact split (proprioception lost on the SAME side as the lesion, pain/temp lost on the OPPOSITE side) is the signature of Brown-Séquard hemisection — here specifically a LEFT-sided lesion (ipsilateral dorsal column = left proprioception loss; contralateral spinothalamic = right pain/temp loss).

Q52. Which tract lesion classically gives a positive Romberg test (patient sways/falls with eyes closed but is stable with eyes open)?

✓ **Correct: B. Dorsal column-medial lemniscus pathway (proprioception loss compensated visually)**

A positive Romberg sign specifically indicates a SENSORY (proprioceptive, i.e., dorsal column) ataxia — vision compensates for the lost proprioceptive input, so the patient is stable with eyes open but becomes unsteady once that visual compensation is removed (eyes closed).

✗ *D is wrong: Cerebellar (truly ataxic) patients are UNSTEADY REGARDLESS of whether their eyes are open or closed — that's a Romberg-negative pattern, and distinguishing a positive from negative Romberg is exactly how you differentiate sensory ataxia (dorsal column) from cerebellar ataxia.*

TIP: Positive Romberg = worse with eyes closed = sensory (dorsal column) problem, vision was compensating. Unsteady with eyes OPEN too = NOT a true positive Romberg, points toward cerebellar disease instead. This distinction is tested constantly and often worded in tricky, negatively-phrased ways.

Q53. Which of the following conditions primarily affects the PNS rather than the CNS?

✓ **Correct: B. Guillain-Barré syndrome**

Guillain-Barré syndrome is an autoimmune, ascending demyelinating polyneuropathy that targets PERIPHERAL nerve myelin (Schwann cells) — the other options are all primarily CNS diseases.

Q54. A patient has right-sided pain and temperature loss below T12, and left-sided pain and temperature loss below T10 (staggered levels)....

✓ **Correct: B. An expanding syrinx hitting crossing fibers unevenly**

Because spinothalamic fibers cross close to their level of entry, an asymmetric or expanding central lesion (like a syrinx) can catch decussating fibers at DIFFERENT levels on each side as it grows, producing this classic 'staggered' sensory level pattern rather than one clean, single transverse level.

TIP: A 'staggered' or 'dissociated, asymmetric level' sensory pattern is a strong clue for syringomyelia — the crossing spinothalamic fibers right next to the expanding central cavity get hit unevenly on each side as the syrinx grows, whereas a discrete traumatic hemisection tends to give one clean, sharp level.

Q55. The ganglia housing first-order taste neurons are located in which structures?

✓ **Correct: A. Geniculate, petrosal, nodose (CN VII, IX, X)**

Taste afferents from the anterior 2/3 tongue (CN VII, geniculate ganglion), posterior 1/3 tongue (CN IX, petrosal/inferior ganglion), and epiglottis/pharynx (CN X, nodose/inferior ganglion) all carry their first-order cell bodies in these three respective ganglia.

TIP: Taste = 3 cranial nerves, 3 ganglia, 3 tongue regions: VII/geniculate = anterior 2/3, IX/petrosal = posterior 1/3, X/nodose = epiglottis/pharynx. Memorize this trio as a set — questions love to ask about any one piece in isolation.

Q56. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?

✓ **Correct: C. Basal cells**

Basal cells are the shared regenerative stem cell population for both taste bud cells and olfactory receptor neurons — a nice unifying fact across two different special senses.

Q57. The gustatory (taste) pathway includes all of the following EXCEPT:

✓ **Correct: E. Trapezoid body**

The trapezoid body is part of the AUDITORY pathway (a brainstem relay in the ascending auditory pathway), not the gustatory pathway — a structure-swap trap between two unrelated special sense pathways.

TIP: Full gustatory pathway ladder: Taste bud → CN VII/IX/X → Geniculate/petrosal/nodose ganglion → Solitary nucleus (rostral part) → VPM thalamus → Insula/frontal operculum (gustatory cortex). Five steps, memorize the whole chain, not just pieces.

Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

Q58. A woman presents with depressive symptoms and suicidal thoughts. Which neurotransmitter is classically implicated as abnormal in this case?

✓ **Correct: C. Serotonin**

Serotonergic dysfunction is the classic neurotransmitter abnormality implicated in major depression, which is why SSRIs are first-line pharmacotherapy.

✗ E is wrong: Dopamine is more classically linked to reward/motivation/psychosis and Parkinson's disease — a tempting but less specific answer for a straightforward depression vignette.

Q59. What is the mechanism of action of botulinum toxin?

✓ **Correct: C. Cleaves SNARE proteins, blocking vesicle fusion**

Botulinum toxin is a protease that cleaves SNARE proteins (e.g., SNAP-25, syntaxin, VAMP) required for vesicle docking/fusion, directly preventing ACh release regardless of whether Ca^{2+} has entered the terminal.

✗ D is wrong: Blocking Ca^{2+} channels would indirectly reduce release (this is closer to Lambert-Eaton's autoimmune mechanism), but botulinum's actual target is the SNARE fusion machinery ITSELF, downstream of Ca^{2+} entry — a subtle but important mechanistic distinction.

Q60. An 80-year-old man has progressive short-term memory loss and word-finding difficulty, diagnosed with Alzheimer disease. Which statement...

✓ **Correct: E. Blocking of dopamine reuptake transporters**

Dopamine reuptake blockade has nothing to do with Alzheimer's pathogenesis — that's a completely different neurotransmitter system, more relevant to Parkinson's disease or stimulant pharmacology, making it the clearly false statement.

TIP: Alzheimer's core pathology triad: amyloid-beta plaques (extracellular) + neurofibrillary tangles (intracellular, hyperphosphorylated tau) + cortical atrophy (especially hippocampus/temporal-parietal). Beta-secretase (BACE1) is the enzyme that cleaves APP toward the amyloidogenic pathway — increased activity worsens plaque formation.

Q61. A patient arrives with continuous generalized tonic-clonic seizures for 15 minutes (status epilepticus). What is the most appropriate...

✓ **Correct: A. Diazepam, a benzodiazepine**

Benzodiazepines (IV lorazepam or diazepam) are first-line for actively seizing status epilepticus because of their rapid onset via GABA-A potentiation; second-line agents (phenytoin, valproate, levetiracetam) are used if seizures persist.

✗ D is wrong: Fluoxetine (an SSRI antidepressant) has no role in acute seizure termination — an obviously wrong-category distractor testing whether you're just pattern-matching to any 'drug name' rather than the actual clinical indication.

TIP: Status epilepticus drug ladder: Benzodiazepine FIRST (fast GABA-A potentiation) → then phenytoin/fosphenytoin or valproate/levetiracetam SECOND-line if seizures continue → then general anesthesia (e.g., propofol) as a last resort for refractory status.

Q62. The summation of EPSPs arriving from numerous different presynaptic fibers converging onto the same postsynaptic neuron at roughly the...

✓ **Correct: A. Spatial summation**

Multiple different synapses (different locations = spatial) firing together add their depolarizations — spatial summation.

✗ *B is wrong: Temporal summation is the SAME synapse firing repeatedly in rapid succession, a time-based (not location-based) mechanism — the question specifically describes 'numerous fibers,' which is a spatial, not temporal, scenario.*

Q63. A 20-year-old college student has depressive symptoms including sleep disturbance and anorexia for 1 month, with no prior mood disorder...

✓ **Correct: C. Inhibition of the serotonin transporter (SERT)**

SSRIs (selective serotonin reuptake inhibitors) work by blocking SERT, the presynaptic serotonin reuptake transporter, leaving more serotonin available in the synaptic cleft over time.

Q64. A woman presents with progressive muscle weakness, dysphagia, and ptosis that worsens with repeated blinking. Which finding is most...

✓ **Correct: C. Anti-nicotinic ACh receptor antibody**

Fatigable weakness worsening with repeated activity (repeated blinking causing ptosis to worsen) is classic myasthenia gravis, caused by autoantibodies against postsynaptic nicotinic ACh receptors.

✗ *B is wrong: Lambert-Eaton syndrome antibodies target presynaptic Ca²⁺ channels and classically cause weakness that IMPROVES with repeated activity — the opposite pattern from this vignette's worsening-with-use presentation.*

Q65. A 45-year-old man has a deep gardening wound, then develops jaw stiffness (lockjaw) and muscle spasms. Which mechanism best explains the...

✓ **Correct: D. Blocks inhibitory interneurons in the cord**

This is tetanus (*Clostridium tetani*), not botulism — tetanospasmin travels retrogradely to the CNS and specifically cleaves SNARE proteins in INHIBITORY (glycinergic/GABAergic) interneurons, removing normal inhibition of motor neurons and causing sustained muscle rigidity/spasm — the opposite clinical picture from botulinum toxin's flaccid paralysis, despite a similar SNARE-cleaving mechanism.

TIP: Botulinum and tetanus toxins both cleave SNARE proteins by a nearly identical molecular mechanism, but they travel to and act on completely different neurons — botulinum stays at the NMJ (blocking excitatory ACh release → flaccid paralysis), while tetanus toxin travels retrogradely up the axon into the CNS to block INHIBITORY interneurons (→ spastic rigidity/spasm). Same weapon, opposite target, opposite clinical picture — one of the best 'contrast pair' questions in all of pharmacology.

Q66. A chemotherapy patient has nausea and vomiting. Which antiemetic receptor target is most appropriate?

✓ **Correct: A. 5-HT₃ receptor, e.g. ondansetron**

Chemotherapy-induced nausea/vomiting is strongly mediated via 5-HT₃ receptors in the gut/area postrema; 5-HT₃ antagonists like ondansetron are first-line antiemetics for this specific cause.

Q67. Which of the following agents does NOT stimulate nociceptors (i.e., is not itself an inflammatory/pain-sensitizing mediator)?

✓ **Correct: E. Vibration**

Vibration is a MECHANICAL stimulus detected by mechanoreceptors (like Pacinian corpuscles), not a chemical inflammatory mediator that sensitizes/activates nociceptors — the other four are all classic chemical mediators of the 'inflammatory soup' that sensitizes pain fibers.

TIP: The 'inflammatory soup' sensitizing nociceptors: prostaglandins, bradykinin, K⁺, H⁺, histamine, substance P, ATP — memorize this list as a set, since 'which of these does NOT belong' questions are extremely common.

Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

Q68. A patient exhibits a near-vision reflex defect. Which of the following is NOT a normal characteristic of the near vision reflex?

✓ **Correct: B. Levator palpebrae contraction, lid elevation**

The near reflex TRIAD is convergence + pupillary constriction + accommodation (increased lens convexity) — eyelid elevation (levator palpebrae) is not part of this triad at all, making it the correct 'NOT characteristic' answer.

TIP: Near reflex triad = 3 C's roughly: Convergence, Constriction (pupil), (increased) Convexity of lens. If an option mentions eyelids, that's an automatic red flag — eyelid function is not part of the near triad.

Q69. A patient has a tumor compressing the optic chiasm. Which visual field defect results?

✓ **Correct: C. Bitemporal heteronymous hemianopia**

The optic chiasm is where NASAL retinal fibers (carrying TEMPORAL visual field information) cross; compressing the chiasm classically damages these crossing fibers bilaterally, causing loss of both temporal visual fields — bitemporal hemianopia (classic pituitary tumor presentation).

TIP: Optic chiasm compression (classically from a pituitary adenoma pressing upward) = bitemporal hemianopia, the single most tested visual field defect in all of neuroanatomy. 'Tunnel vision' from losing both peripheral fields is the classic clinical description.

Q70. Which part of the retina has NO photoreceptor cells at all?

✓ **Correct: D. Optic disc, the physiologic blind spot**

The optic disc is purely the exit point for ganglion cell axons/central vessels, with zero photoreceptors — the anatomical basis of the physiologic blind spot.

Q71. What structure in the eye is primarily responsible for absorbing stray light, reducing internal reflection/glare?

✓ **Correct: A. The pigmented choroid layer**

The heavily pigmented choroid absorbs stray light that passes through the retina, preventing internal scatter/reflection that would otherwise degrade image quality — functioning like the black interior of a camera.

Q72. Which condition is indicated by an intraocular pressure (IOP) greater than 25 mmHg?

✓ **Correct: D. Glaucoma**

Elevated IOP (normal roughly 10-21 mmHg) is the hallmark of glaucoma, which can progressively damage the optic nerve (via mechanical/ischemic injury to ganglion cell axons) if untreated.

Q73. Which structure is found at the level of the rostral pons?

✓ **Correct: D. Motor and principal nuclei of CN V**

The trigeminal motor and principal sensory nuclei sit at the level of the rostral (mid-upper) pons — a classic internal brainstem landmark question.

X B is wrong: The abducens nucleus is at the level of the CAUDAL pons (with the facial colliculus), one level more caudal than this question is asking about.

Q74. A patient has facial nerve damage resulting in hyperacusis (abnormally loud perception of sound). Which structure's function is most...

✓ **Correct: D. Stapedius reflex loss, CN VII lesion**

The stapedius muscle (dampening ossicle movement in response to loud sound) is innervated by the facial nerve; facial nerve damage disables this protective reflex, causing hyperacusis (sounds seem abnormally loud).

X C is wrong: The tensor tympani is innervated by CN V (trigeminal), not CN VII — a facial nerve lesion would not directly affect this muscle's reflex.

TIP: Two ossicle-dampening muscles, two different cranial nerves: Tensor tympani = CN V. Stapedius = CN VII. A facial nerve lesion knocks out only the stapedius reflex, producing hyperacusis specifically — a great example of how remembering WHICH nerve innervates WHICH muscle lets you predict the exact clinical finding.

Q75. Which vestibular nucleus sends axons directly to the oculomotor nucleus (contributing to the vestibulo-ocular reflex)?

✓ **Correct: C. Superior vestibular nucleus**

The superior vestibular nucleus is the primary source of ascending vestibulo-ocular reflex connections (via the MLF) to the oculomotor and other extraocular motor nuclei, coordinating eye movements with head movements.

✗ *B is wrong: The lateral vestibular nucleus (Deiters' nucleus) instead gives rise to the lateral vestibulospinal tract, a DESCENDING pathway to the spinal cord for postural/antigravity muscle tone — a different job entirely from the ascending oculomotor connections.*

Q76. Which structure releases (dissipates) the pressure wave of sound energy out of the inner ear, allowing continued fluid movement within...

✓ **Correct: C. Round window**

The round window (a flexible membrane at the base of the scala tympani) bulges outward to release the pressure wave transmitted through the cochlear fluids, allowing continuous fluid movement despite the fluid's incompressibility — without it, the oval window's inward pushes would have nowhere to 'go.'

✗ *B is wrong: The oval window is the ENTRY point (where the stapes footplate pushes IN), the exact opposite role from the round window's pressure-RELEASE function — these two windows are constantly swapped in student answers.*

TIP: Oval window = IN (stapes pushes fluid in). Round window = OUT (releases pressure, bulges outward). Sound energy flow: air → tympanic membrane → ossicles → oval window → cochlear fluid wave → round window releases the pressure.

Q77. Which statement regarding sound stimulation and cochlear transduction is TRUE?

✓ **Correct: C. Inner hair cells signal via spiral ganglion fibers**

Inner hair cell stereocilia deflection (relative to the tectorial membrane, as the basilar membrane moves) generates the receptor potential that ultimately triggers action potentials in the peripheral processes of spiral ganglion neurons — the actual primary afferent pathway for hearing.

✗ *D is wrong: High frequencies are encoded at the BASE of the cochlea (stiff, narrow basilar membrane), not the apex — the apex is tuned to LOW frequencies, the exact opposite of what this option claims.*